

List of question of slide - 9

1. Draw the arrangement of socket in process to process communication (p-4 fig only)

What is a Socket in Computer Networks?



A **socket** is an **endpoint for sending or receiving data** across a computer network. It is the **interface between the application layer and the transport layer** (like TCP or UDP).



Key Points:

- A socket is defined by:

(IP address, Port number)

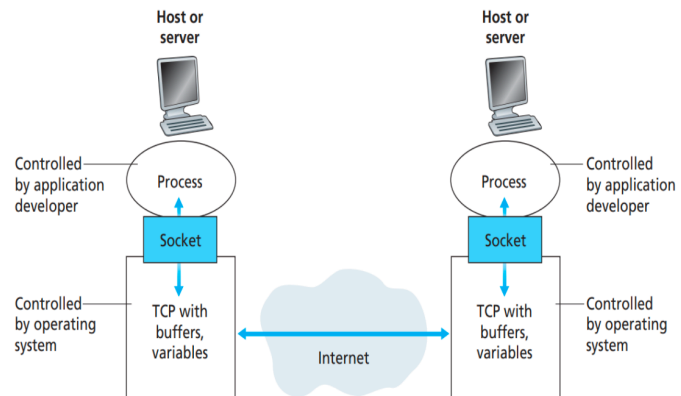
- It enables communication between:
 - **Processes on the same device**, or
 - **Processes on different devices** over the internet or local network
- Sockets are used in **client-server communication**:
 - The **server** creates a socket to listen for incoming connections.
 - The **client** uses a socket to connect to the server.



Types of Sockets:

Type	Protocol
Stream Socket	TCP
Datagram Socket	UDP

- ✓ Consider an analogy to help us understand processes and sockets. A process is analogous to a house and its socket is analogous to its door.



- ✓ When a process wants to send a message to another process on another host, it shoves (or push) the message out its door (socket). This sending process assumes that there is a transportation infrastructure from sending door to the door of the destination process.
- ✓ Once the message arrives at the destination host, the message passes through the receiving process's door (socket), and the receiving process then acts on the message.

2. How a process is addressed in a host?(p-5)

A process running on sending host needs the full address of the process running on the receiving host.

To identify each process, two pieces of information is needed :

1. 🧩 IP Address (Internet Protocol Address)

- **Identifies the host (machine)** in a network.
- It's like the **street address** of a house — tells the network *where* to find the device.

2. 🧩 Port Number

- **Identifies the specific process or application** running on that host.
- It's like the **apartment number** — tells the OS *which process* to deliver data to.
- Each application or service listens on a **specific port**.

This port number is needed because in general a host runs many network applications.

Without a port number, the operating system would not know **which application** should receive the incoming data.

✓ For example, a Web server is identified by port number 80. A mail server process (using the SMTP protocol) is identified by port number 25.

3. Give the idea of net app and app protocol

✓ An application layer protocol is associated with each application. Consider World Wide Web browsers that are available (Firefox, Safari, Internet Explorer etc.). You can use any one of these application programs to access a particular site on the Web is because they all use the same application layer protocol: HTTP (Hyper Text Transport Protocol).

4. Why DNS server is used?

Why is a DNS Server Used?

A **DNS (Domain Name System) server** is used to **translate human-readable domain names** (like www.google.com) into **IP addresses** (like 142.250.190.68) that computers use to identify each other on the network.

Key Reasons DNS is Used:

1. Name-to-IP Translation

- Users type domain names, not IP addresses.
- DNS converts the domain into an IP address so browsers can locate the server.

2. Simplifies Internet Usage

- It's easier to remember www.facebook.com than 157.240.229.35.

3. Scalability

- DNS is a distributed system, making it scalable across the entire internet.

4. Load Distribution

- DNS can map a single domain to multiple IPs for **load balancing**.

5. Redundancy and Reliability

- Multiple DNS servers ensure continuous service even if one fails.

💡 Analogy:

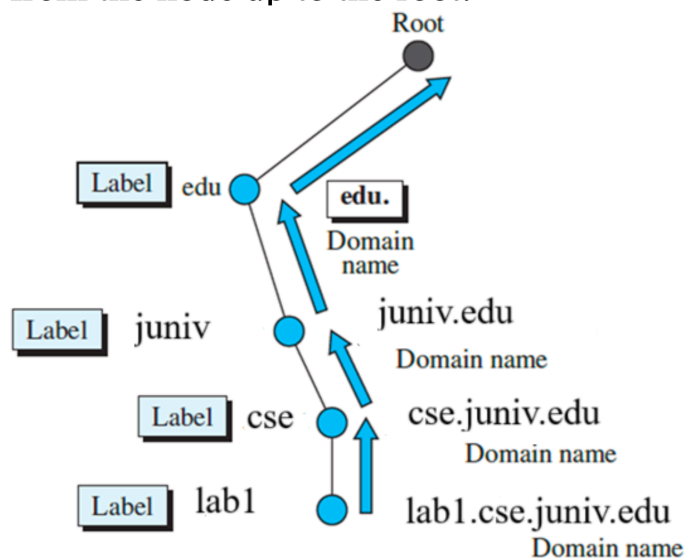
Think of DNS as the **phonebook of the internet**—you look up a name (domain), and it gives you the number (IP address).

Let me know if you'd like to see how a DNS query works step-by-step!

5. Draw the hierarchy of Domain Name and level (fig, no des)

Domain Name

Each node in the tree has a domain name. A full **domain name** is a sequence of labels separated by dots (.). The domain names are always read from the node up to the root.

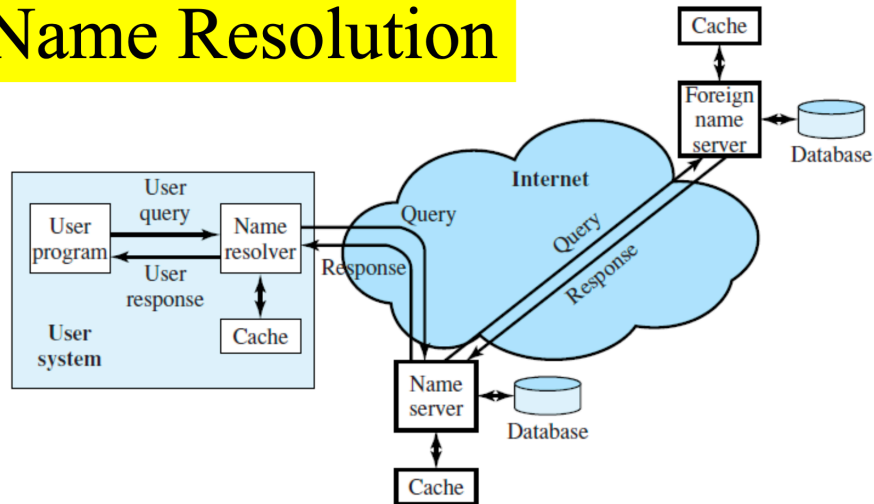


Domain names and labels

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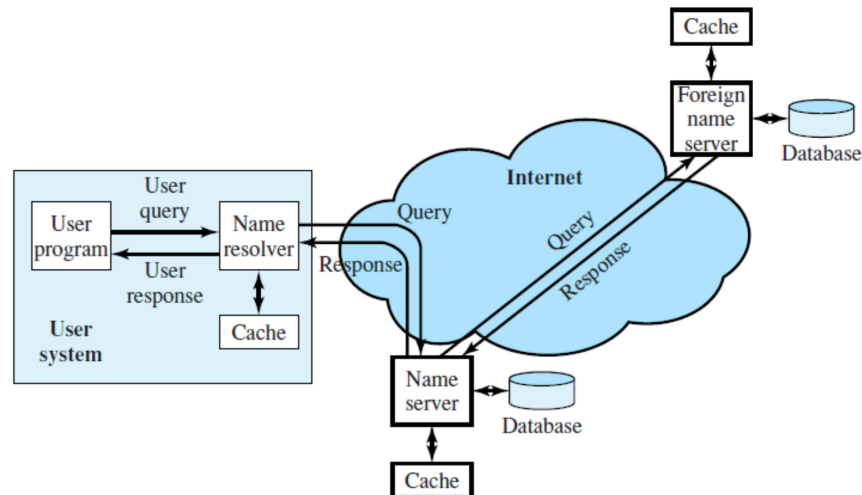
6. Give the steps DNS name resolution.

DNS Name Resolution



DNS operation typically includes the following steps (Figure above):

1. A user program requests an IP address for a domain name.
2. The local name server checks to see if the name is in its local database or cache, and, if so, returns the IP address to the requestor.
3. Otherwise, the name server queries other available name servers, starting down from the DNS tree or as high up the tree as possible.



4. When a response is received at the local name server, it stores the name/address mapping in its local cache and may maintain this entry for the amount of time specified in the **time to live** field of the retrieved RR (*resource records*).
5. The user program is given the IP address or an error message.

7. Mention the function of five tuple of DNS record (no table)

✓ The Five Fields of a DNS Resource Record:

Field	Function
1. Domain_name	Specifies the domain (or subdomain) to which the record belongs.
2. Time_to_live (TTL)	Indicates how long the record can be cached by resolvers (in seconds).
3. Class	Specifies the protocol family . Usually IN for Internet records.
4. Type	Describes the kind of record , such as A, MX, CNAME, etc.
5. Value	Contains the actual data for the record. Can be an IP, hostname, etc.

📌 Example:

www.example.com. 3600 IN A 93.184.216.34

- **Domain_name:** www.example.com
- **TTL:** 3600 seconds (1 hour)
- **Class:** IN (Internet)
- **Type:** A (Address record)
- **Value:** 93.184.216.34 (IP address)

Let me know if you want a table of the most common DNS record types and their purposes.

<i>Type</i>	<i>Interpretation of value</i>
A	A 32-bit IPv4 address (see Chapter 7)
NS	Identifies the authoritative servers for a zone
CNAME	Defines an alias for the official name of a host
SOA	Marks the beginning of a zone
MX	Redirects mail to a mail server
AAAA	An IPv6 address (see Chapter 7)

The principal DNS resource record types

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8. Example-1 What do you mean by record:

www.ait.ac.th 86400 IN. A 192.168.30.44

Ans.

Meaning:

- The **domain name** www.ait.ac.th
- Has a **Time To Live (TTL)** of 86400 seconds (i.e., 24 hours)
- Belongs to the **Internet class** (IN)
- Is of **Type A**, meaning it maps to an **IPv4 address**
- The **value** or IP address is 192.168.30.44

-

DNS server will provide IP address 192.168.30.44 against
www.ait.ac.th

9. Example-2 What do you mean by record:

✓ Example:

Record: ait.ac.th 86400 IN MX kuddus.ait.ac.th

✓ Meaning:

- **Domain name:** ait.ac.th
 - **TTL (Time To Live):** 86400 seconds (24 hours)
 - **Class:** IN (Internet)
 - **Type:** MX (Mail Exchange)
 - **Value:** kuddus.ait.ac.th (mail server responsible for receiving emails)
-

✓ Interpretation:

This record tells us that **emails sent to any user at @ait.ac.th** (e.g., bill@ait.ac.th) should be **routed to the mail server** named kuddus.ait.ac.th.

The sender's mail system will query DNS to get the MX record for ait.ac.th and deliver the message to the specified mail server.

10. How a mail box is addressed under SMTP

- In order for the mail to be sent properly from mailbox to mailbox, each mailbox must have a unique e-mail address.
- The e-mail address is multipart hierarchical address. The first portion of the address indicates the user and the later portion indicates the location of the mailbox.
- A typical message starts its journey in the sender's user agent, travels to the sender's mail server, and travels to the recipient's mail server, where it is deposited in the recipient's mailbox.

11. Mention the function of 3 phases of POP3 (Post Office Protocol v3)

- During the first phase, **authorization**,
 - the user agent sends a username and a password (in the clear) to authenticate the user.
- During the second phase, **transaction**,
 - the user agent retrieves messages;
 - also during this phase, the user agent can mark messages for deletion,
 - remove deletion marks,
 - and obtain mail statistics.
- The third phase, **update**,
 - occurs after the client has issued the quit command,
 - ending the POP3 session;
 - at this time, the mail server deletes the messages that were marked for deletion.

13. What are the benefits of web caching

Caching provides four benefits:

- a. Reduce load on the web server for example during admission test of Jahangir University.
- b. Reduce latency
- c. Reduce bandwidth utilization of users
- d. Network traffic reduction

13. Why Network Management Software is necessary (P':70)

- The main purpose of network management is to monitor, manage, and control a network. A network can be structured with many links, routers, switches, servers, and other physical-layer devices, which can be equipped with many network protocols that coordinate them.

- Imagine when thousands of such devices or protocols are tied together by an ISP/organization network and how drastic their management can become to avoid any interruptions in routine services.
- In this context the purpose of network management is to monitor, test, and analyze the hardware and software of components of the entire network.